SECTIONS WITH SUPPORT IN A LOCALLY CLOSED SUBSCHEME ARE INDEPENDENT OF THE OPEN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Sections with support in a closed subscheme of an open (more-etale, Lemma lemma-section-support-in-locally-closed-pre, canonical tag 0F6F). cited at proof line 309. Gives the canonical bijection in the case U' subset U, of which the statement is a special case.

Source reference. Original lemma 0F6F: sources/stacks-chow/tags.txt, source lines 18970–18970. Cited at proof line 309: the case U' subset U is its special case.

Lemma 0.2. *Let X be a scheme. Let $Z \subset X$ be a locally closed subscheme. Let $\mathcal{F}$ be an abelian sheaf on $X_{étale}$. Given $U, U' \subset X$ open containing Z as a closed subscheme, there is a canonical bijection*

$$\{s \in \mathcal{F}(U) \mid Supp(s) \subset Z\} = \{s \in \mathcal{F}(U') \mid Supp(s) \subset Z\}$$

which is given by restriction if $U' \subset U$.

Proof. Since Z is a closed subscheme of $U \cap U'$, it suffices to prove the lemma when $U' \subset U$. Then it is a special case of Lemma 0F6F. □

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